

HRFlow

The HRM

HR Management System with Integrated Interview Management System

Project Overview Document

A comprehensive web-based platform for managing the complete employee lifecycle — from recruitment and interviews to attendance, projects, performance, and AI-powered decision support.

Project Overview

The HR Management System (HRMS) is a comprehensive web-based platform designed to manage the complete employee lifecycle within an organization. It centralizes human resource operations, recruitment, interview management, employee administration, attendance, project assignments, performance monitoring, and AI-powered decision support into a single application.

Instead of using multiple tools for recruitment, attendance, employee records, and project management, the HRMS provides one unified platform where HR managers, interviewers, employees, and administrators can perform all their daily tasks efficiently.

One of the core modules of the system is the Interview Management System (IMS), which automates the recruitment process from job posting to candidate hiring. Once a candidate is selected, the system seamlessly converts them into an employee, allowing HR to continue managing them within the same platform.

Main Objectives

- Digitize and automate HR operations.
- Streamline the recruitment and interview process.
- Maintain centralized employee records.
- Track attendance and working hours.
- Manage company projects and employee tasks.
- Automatically screen and rank candidate CVs using AI.
- Assist HR managers in allocating employees to projects based on skills and workload.
- Generate real-time reports and analytics for management.

Core Modules

1. Human Resource Management (Main Module)

This is the heart of the system. It manages:

- Employee profiles
- Departments
- Teams
- Attendance
- Leave management (future enhancement)
- Performance monitoring
- Project assignments
- Task management
- HR reports
- Employee analytics

2. Interview Management System

This module handles recruitment.

HR can:

- Create job vacancies
- Configure interview processes
- Build technical quizzes
- Review applications
- Schedule interviews
- Evaluate candidates
- Hire or reject applicants

Candidates can:

- Register
- Apply for positions
- Upload CVs
- Complete assessments
- Attend interviews
- Track application progress

Once hired, candidates are automatically converted into employee accounts.

3. AI Recruitment Module

The AI engine assists HR by:

- Parsing resumes

- Extracting candidate skills
- Comparing CVs with job requirements
- Ranking candidates
- Recommending the most suitable applicants

This significantly reduces manual screening time and improves recruitment quality.

4. Employee Management

HR can manage:

- Personal information
- Employment details
- Departments
- Positions
- Skills
- Documents
- Performance history
- Attendance records

5. Attendance Management

Employees can:

- Clock in
- Clock out

The system records:

- Working hours
- Daily attendance
- Monthly summaries
- Attendance reports

6. Project & Task Management

Managers can:

- Create projects
- Assign employees
- Create tasks
- Monitor progress
- Track deadlines

Employees can update task statuses and view assigned work.

7. AI Resource Allocation

When a new project is created, the AI suggests the most suitable employees based on:

- Skills
- Experience
- Current workload
- Availability

Managers can accept or modify these recommendations.

8. Reports & Analytics

The platform provides dashboards and reports for:

- Employee statistics
- Recruitment performance
- Attendance
- Project progress
- Task completion
- Hiring trends
- Department performance

User Roles

The system includes four primary user roles:

Administrator

- Manage the entire system
- User and role management
- System configuration
- Reports and analytics
- Security and permissions

HR Manager / Interviewer

- Manage recruitment
- Conduct interviews
- Create job postings
- Manage employees
- Track attendance
- Assign projects
- Review AI recommendations

Employee

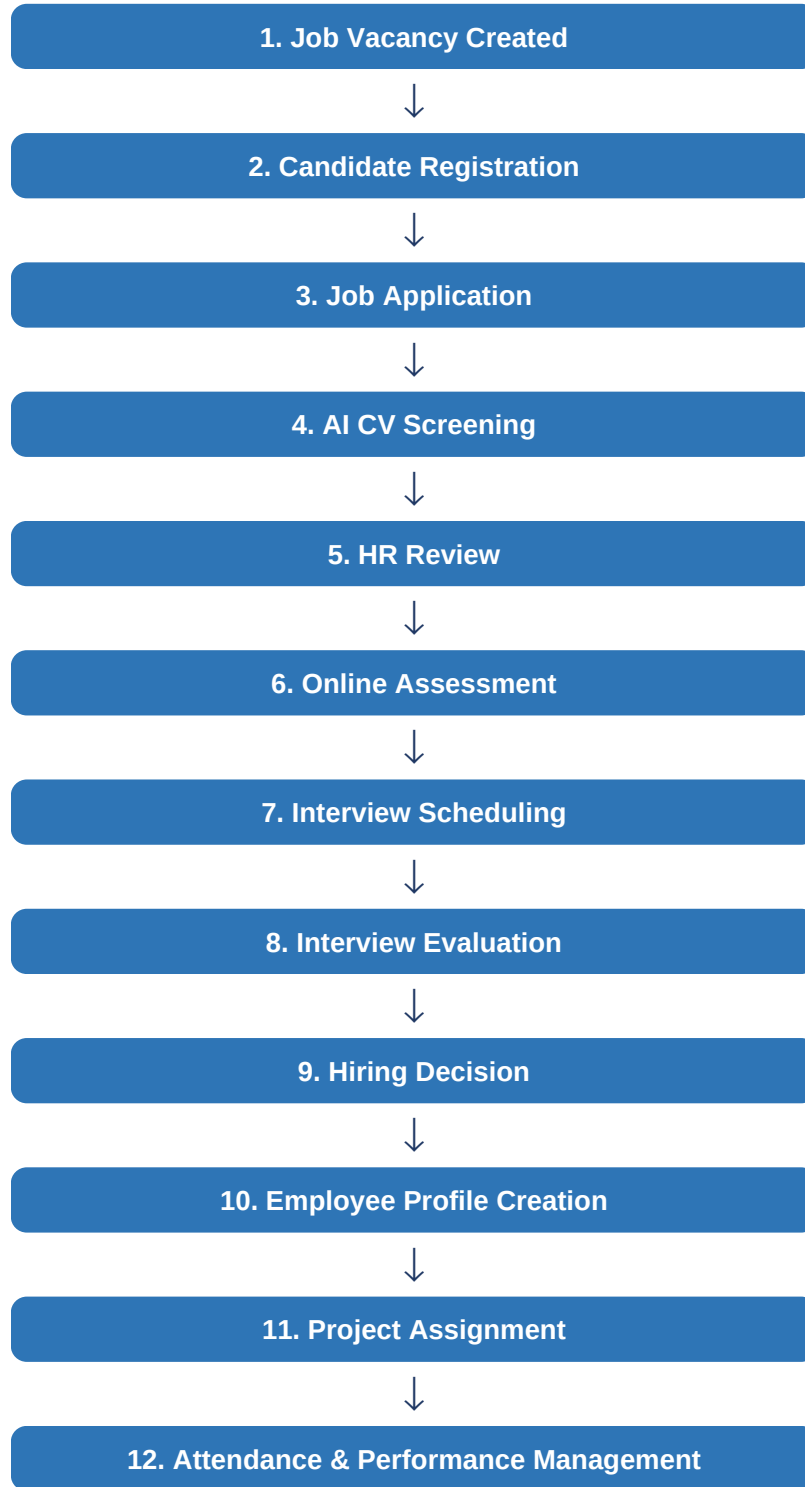
- View assigned projects
- Manage tasks
- Record attendance
- Update profile
- View performance

Candidate

- Register
- Apply for vacancies
- Upload resumes
- Complete assessments
- Attend interviews
- Monitor application status

System Workflow

The recruitment-to-management pipeline follows a clear, linear sequence of stages:



Key Features

- Secure Authentication & Role-Based Access
- Employee Management
- Interview Management
- Recruitment Workflow
- AI Resume Screening
- Online Assessments
- Interview Scheduling
- Attendance Tracking
- Project & Task Management
- AI Resource Allocation
- Notifications
- Reports & Analytics
- Real-Time Updates

Technology Stack

Layer	Technology
Frontend	React.js
Backend	Node.js + Express.js
Database	MongoDB
AI Service	Python (FastAPI)
Authentication	JWT + bcrypt
Real-Time Communication	Socket.io
File Storage	Cloudinary or AWS S3
Video Interviews	Zoom SDK or Jitsi

Final Project Vision

HRFlow should be presented as an Enterprise HR Management System, with the Interview Management System acting as one of its major modules. Recruitment is the starting point, but the platform's primary purpose is to manage employees throughout their entire lifecycle — from hiring and onboarding to attendance, project management, performance tracking, and workforce optimization.

This positioning makes the system suitable for medium to large organizations and aligns with enterprise HR software solutions.